MATH 4350 Notes - Number Theory and Cryptography

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Contents

1 Syllabus / How to Class Works

- See course website: https://www.math.wustl.edu/ matkerr/4350/

2 Introduction

- **GOAL:** How do we disguise messages s.t. only the sender and receiver know what the message is even when the encryption strategy is known?

2.1 Greatest Common Divisors

- Let $\mathbb{N} = \{1, 2, 3, \dots\}$ be the set of natural numbers.
- Let $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$ be the set of integers.
- **Def.** Greatest Common Divisor (GCD): $(a, b) :=$ largest element of $S(a, b) := \{d \in \mathbb{N} | d \mid a, b\}$
- Properties of the GCD (true, but not proved in this class):
  - $a|b \wedge b|c \Rightarrow a|c$
  - $a|b \wedge b|a = a = \pm b$
  - if $a|b$ and $a|c$ and $x, y \in \mathbb{Z}$, then $a|(bx + cy)$
- Recall that induction exists lmao
- **Ex.** Find the GCD of $(a, b) = (162, 72)$
  - Use GCD props:
  - $(0, a) = a = (a, a)$
  - $(a, b) = (b, a) = (a, -b)$
  - $(a, b - ma) = (a, b)$ for some $m \in \mathbb{Z}$
  - Recall long division: $A = \{a - bk | k \in \mathbb{Z}\, a - bk \geq 0\} \subseteq \mathbb{N} \cup \{0\}$ has least element r if $r \geq b$, then $r - b \geq 0 \Rightarrow r \in A$. Thus $a = bq + r$ for some unique $q, r \in \mathbb{Z}$ and $0 \leq r < b$.
  - Going back to the example,

$$162 = 72 \cdot 2 + 18 \Rightarrow (162, 72) = (18, 72) = (72, 18)$$

$$72 = 18 \cdot 4 + 0 \Rightarrow (72, 18) = (18, 0) = 18$$

  Thus, $(162, 72) = 18$. Let's generalize this:
- **Def.** Euclidean Algorithm:
  - $a = bq_1 + r_1$, $0 \leq r_1 < b$
  - $b = r_1 q_2 + r_2$, $0 \leq r_2 < r_1$
  - $r_1 = r_2 q_3 + r_3$, $0 \leq r_3 < r_2$
  - $\vdots$
  - $r_{n-1} = r_n q_{n+1} + r_{n+1}$
  - Eventually, $r_{n+1} = 0$, so $r_{n-1} = r_n q_{n+1} + 0$
  - Thus, $(a, b) = r_n$
  - Eventually, we will prove this is a polynomial time algorithm.
- **Thm.** $S := \{ax + by | x, y \in \mathbb{Z}, ax + by > 0\} = (a, b) = g$ (i.e. the GCD)

  Proof. First, we must show $S \subseteq g$. We know that $a = g\alpha$ and $b = g\beta$ since $g|a, b$ by defn for some $x, y \in \mathbb{Z}$.

  Thus, any $ax + by = g(\alpha x + \beta y)$ is divisible by g.

  Next, we must prove that $g \subseteq S$. We know that $g = (a, b) = r_n =$ the last non-zero remainder in the Euclidean Algorithm. We will show all non-zero r_j's are in S via induction:

Base Case: $r_1 = 1 \cdot a + (-q_1) \cdot b \in S$
Inductive Hypothesis: Assume $r_j = a \cdot x_j + b \cdot y_j \in S \quad \forall j \in [1, \cdots, k-1]$.
Inductive Step: We know:

$$r_{k-2} = r_{k-1}q_k + r_k \Rightarrow r_k = r_{k-2} + (-q_k)r_{k-1}$$

$$= a \cdot x_{k-2} + b \cdot y_{k-2} + (-q_k)(a \cdot x_{k-1} + b \cdot y_{k-1}) = a(x_{k-2} - q_k x_{k-1}) + b(y_{k-2} - q_k y_{k-1}) \in S$$

□

2.2 Fermat's Little Theorem

- **Thm.** $a^p \equiv a \mod p$ for all $a \in \mathbb{Z}$ and prime p

  Proof. **Claim:** □